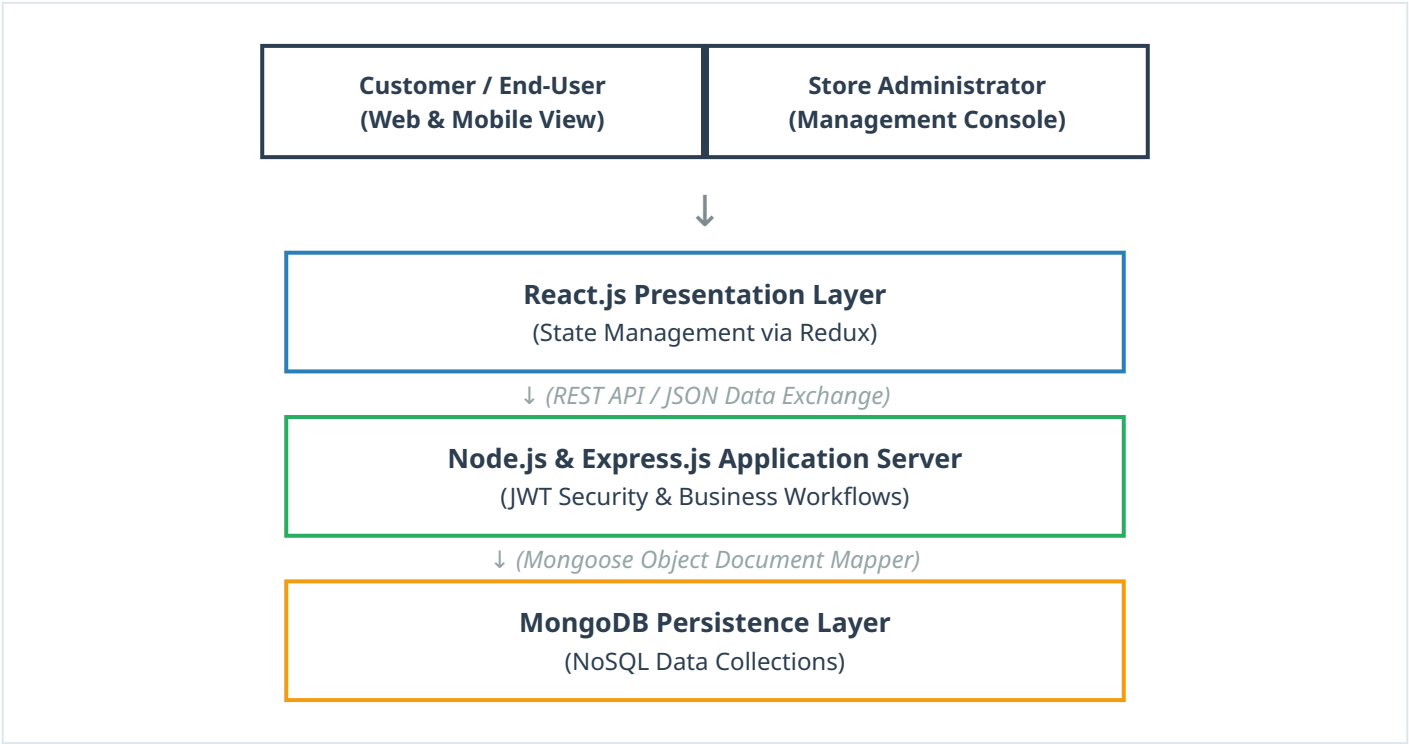


PROJECT DESIGN PHASE-II

Technology Stack & System Architecture

Date	18 June 2026	Project Name	ShopEz (MERN E-Commerce)
Team ID	--	Maximum Marks	4 Marks

1. Technical Architecture (MERN Stack)



2. Components & Technology Stack

Component	Description	Technology
User Interface	Responsive Single Page Application (SPA) providing interfaces for both consumers and staff.	React.js, Tailwind CSS
Backend Engine	Facilitates REST interface, user authorization, and core application logic.	Node.js & Express.js
Core Database	Distributed NoSQL storage for managing profiles, products, and order data.	MongoDB Atlas
Identity Management	Token-based session control combined with robust password hashing algorithms.	JWT & Bcrypt.js
State Management	Centralized synchronization of shopping basket and logged-in user information.	React Context API / Redux

Component	Description	Technology
Hosting Infrastructure	Staging and live deployment environments for high availability.	Local / Vercel / Heroku

3. Module Design & Data Architecture

Module	Functional Description	Key Database Fields
Authentication Module	Manages account creation, identity verification, and secure token issuance.	email, password_hash, role
Inventory Module	Oversees the product library, faceted search, and category classification.	name, unit_price, category, stock_level
Transaction Module	Processes customer orders while maintaining purchase logs and status updates.	customer_id, product_items[], grand_total, status

4. API Interface Design (REST)

Endpoint	HTTP Method	Purpose
/api/users/login	POST	Validates credentials and generates a secure session token.
/api/products	GET	Retrieves catalog data or filtered results based on user preferences.
/api/orders/place	POST	Transmits order specifics for validation and database entry.
/api/admin/product	POST/PUT	Restricted path for staff to add or modify merchant listings.

5. Design Considerations

Scalability: By leveraging MongoDB's NoSQL structure, the system facilitates dynamic schema evolution and efficient horizontal scaling. **Security:** All sensitive endpoints are protected via JWT middleware; credentials are salted and hashed to ensure they are never stored in readable formats. **Performance:** The utilization of React's Virtual DOM ensures rapid UI re-rendering during intensive operations like filtering or updating the shopping cart.